

Oulamuna Project

Below is a detailed requirements document and set of technical recommendations for building a website that honors Algerian scholars by allowing users to create profiles for them, including photos, biographies, publications, and articles.

1/Project Requirements Document:

Project Overview

This project involves creating a website to showcase Algerian scholars, allowing users to create profiles for scholars they admire. The site will include options to add photos, biographies, publications (articles, books), and relevant external links or videos about each scholar. Users should be able to contribute information easily, and an admin dashboard will allow the site owner to validate and manage the content.

Objectives

1. **User-Generated Profiles:** Enable users to create and edit profiles for Algerian scholars, adding media and content.
2. **Administrative Control:** A secure dashboard for administrators to review, validate, and manage user-submitted content.
3. **Media Support:** Allow users to upload PDFs, link to videos, and display relevant media directly on the scholar profiles.
4. **User-Friendly Design:** Provide an intuitive, engaging, and aesthetically pleasing interface.
5. **Scalability:** Build a scalable back-end capable of handling high volumes of media and user data.

Functional Requirements

User Interface (Front-End)

- User Registration and Login: Allow users to create accounts to submit profiles. Implement Google/Facebook authentication for convenience.
- Profile Creation and Submission: Form for users to enter:
 - Scholar's name, photo, and short biography.
 - Detailed sections for works, publications (PDF upload), and related articles.
 - Links to external videos or hosted media.
- Scholar Profiles: Each profile page should include:
 - Scholar's image and biography.
 - Sections for books, articles, and related videos or media.

- Links to PDF files or external video sources (embedded where possible).
- Search and Filtering: Search functionality to find scholars by name, field, or keywords. Filters by date, type of work (book, article), etc.
- User Contribution: Users can suggest edits to existing profiles, subject to admin approval.

Admin Dashboard (Back-End)

- Content Management: Ability for the admin to:
 - Review, approve, or reject user-submitted profiles and edits.
- Edit or correct any profile details.
- Remove inappropriate content.
- Analytics and Reporting: Overview of user activity, number of submissions, active users, etc.
- User Management: Admin should have options to ban users or restrict access if necessary.

Technical Specifications

Design and UX

1. Simple and Intuitive Layout: Prioritize ease of use, with clearly marked fields and easy navigation.
2. Color Scheme and Fonts: Use a minimalist design with colors that reflect knowledge and Algerian heritage—neutral colors with accents of green and red.
3. Responsive Design: Ensure the site is optimized for both desktop and mobile devices.
4. Media Presentation: Embed PDF viewers and video players directly within the profile pages for smoother content consumption.

User Experience Considerations

1. User Guidance: Provide tooltips or inline explanations in forms to guide users in filling out details for scholars.
2. Feedback Mechanism: Give users feedback after submitting profiles, e.g., "Profile submitted for review," or show error messages if fields are incorrectly filled.
3. Consistent Design Language: Use a coherent design across pages, with a recognizable iconography for different types of media (books, articles, videos).

Front-End Development

Technologies: Use React.js or Vue.js for a dynamic, responsive interface with reusable components.

1. **Forms and Input Validation:** Use libraries such as Formik or Yup to validate input fields, ensuring data quality before submission.
2. **Media Display:** Embed PDF files with libraries like PDF.js or an iframe viewer; for videos, consider React-Player to support YouTube, Vimeo, and local video hosting.
3. **Authentication:** Use Firebase Authentication for user sign-in with social media options for convenience.
4. **Real-Time Updates:** Consider React Query or SWR for caching and synchronization of data with the back-end to provide a seamless experience.

Back-End Development

Technologies: Use Node.js with Express for scalable server architecture, paired with MongoDB for a NoSQL solution that can handle flexible and large datasets like user-uploaded content.

1. **User Authentication:** Store user accounts securely and enable admin verification roles using JWT (JSON Web Token).
2. **Data Management:** Structure MongoDB collections to separate user data from scholar profiles and submissions, ensuring optimized retrieval and filtering.
3. **File Storage:** Use AWS S3 or Google Cloud Storage for storing PDF files and video links. For cost-efficiency, prioritize linking to hosted videos (e.g., YouTube).
4. **Admin Verification Workflow:** Implement a review queue where user-submitted profiles are held until the admin approves them.

Security and Performance

1. **Input Validation and Sanitization:** Protect against SQL injection, XSS, and other vulnerabilities by thoroughly validating all input.
2. **Scalability with Caching:** Use Redis caching for frequently accessed scholar profiles or search results to reduce database load.
3. **Content Delivery Network (CDN):** Implement a CDN to speed up content delivery, particularly for media-heavy pages.

Additional Recommendations

Testing and Quality Assurance

- Conduct Unit Testing (e.g., using Jest for front-end and Mocha for back-end) to ensure functionality at the component level.
- Perform Usability Testing with a few users before launch to identify any issues in user interaction.
- Load Testing for server performance under high traffic and media uploads.

Deployment and Hosting

- Host the website on a scalable platform like AWS or Google Cloud.
- Use Docker for containerizing the application, allowing seamless deployment across different environments.
- Set up Continuous Integration/Continuous Deployment (CI/CD) pipelines for code updates.

Conclusion

This project requires careful attention to user experience and back-end scalability, especially considering the media storage needs and user-generated content review process. By following these guidelines, the development team can create a robust, user-friendly platform to celebrate and document the contributions of Algerian scholars.